

Power Engineer, Quantitative Modeller & Production Software Builder

Model Evaluation | Uncertainty | Monitoring | Evidence Boundaries | Reproducible Systems

PROFILE

Engineer and quantitative modeller transitioning toward AI safety research, with two TUM master's degrees and more than five years of experience building and validating models, data systems and production software in high-stakes energy environments. Work spans forecasting, statistical validation, backend services, cloud delivery, defect investigation and communication of model limitations. Current independent work on GridPulse focuses on how systems should behave when evidence is incomplete: preserve provenance, quantify uncertainty, expose failure modes and abstain rather than overclaim.

SAFETY-RELEVANT STRENGTHS

Evaluation	Model validation; backtesting; failure analysis; scenario testing; benchmark design; reproducibility
Reliability	Data-quality controls; monitoring; production support; defect investigation; explicit evidence boundaries
Quantitative	Time series; regression; forecasting; uncertainty; market regimes; sensitivity to assumptions
Engineering	Python; C#/ .NET; SQL; PostgreSQL; AWS; Azure; CI/CD; automated testing; analytical interfaces

SELECTED INDEPENDENT WORK

Founder / Technical Lead | GridPulse Insights, Berlin Current

Evidence-aware grid screening and model validation

- Built a public-data Power Finder and backtesting environment for investigating candidate connection locations for large electrical loads.
- Designed provenance, confidence and evidence classes so incomplete data cannot be presented as operator-confirmed grid capacity.
- Implemented scenario-based electrical analysis and reproducible case records; current work tests uncertainty, benchmark coverage and abstention conditions.
- Research question: how should AI-assisted infrastructure tools combine probabilistic inference with deterministic engineering checks without hiding uncertainty?

Product Builder | RegenSync, Berlin Earlier project

Role-aware workflow exploration for regenerative-therapy pipelines

- Explored evidence navigation and role-aware workflow design in a safety-sensitive clinical domain.
- Learned that technical completeness does not substitute for a sharply defined user, decision boundary and validation path.

PROFESSIONAL EXPERIENCE

Energy Market Consultant | Baringa Management, Dusseldorf 11/2024 - 05/2026

Energy technology delivery and production readiness

- Structured requirements, risks and testing across an international energy-IT transformation.
- Coordinated defect resolution, production cutover and stabilisation across technical and operational teams.

Full Stack DevOps / C# Developer | Xapio, Munich 11/2022 - 09/2024

Backend systems, controlled cloud delivery and software quality

- Developed modular .NET services and PostgreSQL workflows with Azure Functions, CI/CD and xUnit testing.
- Investigated production and data-quality issues, strengthening validation, reliability and maintainability.

PROFESSIONAL EXPERIENCE CONTINUED

Quantitative Analyst - Energy Markets | GETEC Energie, Hannover

06/2021 - 10/2022

Forecasting, data validation and decision-facing analytical tools

- Developed Python short-term and day-ahead electricity-price forecasts and gas-price models for TTF and THE.
- Validated time-series data, fundamentals, scenarios and assumptions; translated outputs for analysts and commercial decisions.
- Built a user-facing Python interface and Power BI reporting for model interpretation and adoption.

Quantitative Analyst for Market Models | Likron, Munich

11/2019 - 05/2021

Algorithmic-trading models, AWS data workflows and live-system support

- Developed Python calibration prototypes and C# components for energy-trading and execution systems.
- Built SQL/AWS workflows, tested software behaviour and investigated live data and application failures.

Quantitative Analysis Intern | Statkraft Markets, Dusseldorf

10/2018 - 03/2019

Statistical modelling, assumption testing and documentation

- Developed UK imbalance-price and Net Imbalance Volume models using Python and regression.
- Validated large time-series datasets, model logic and forecast behaviour; documented methods and limitations.

EDUCATION

M.Sc. Management | TUM School of Management

2022 - 2025

Focus: Entrepreneurship, technology assessment and product-oriented decision-making

M.Sc. Power Engineering | Technical University of Munich

2016 - 2019

Focus: Power systems, electricity markets, statistics, renewable integration and quantitative modelling

B.Sc. Mechanical Engineering | Islamic Azad University Mashhad

2010 - 2014

Focus: Energy conversion, renewable energy, thermodynamics and engineering fundamentals

CURRENT AI SAFETY RESEARCH INTERESTS

- Evaluation of tool-using and agentic systems under incomplete, conflicting or distribution-shifted evidence.
- Calibrated uncertainty and abstention: when a system should defer rather than convert weak evidence into a confident action.
- Monitoring and incident-oriented evaluation for AI components embedded in consequential technical workflows.
- Combining learned models with deterministic checks, provenance and reproducible test cases for human-auditable decisions.

RESEARCH PREPARATION

Relevant foundations include statistical modelling, experiment and backtest design, assumption testing, model-to-production delivery, technical documentation and cross-disciplinary communication. I have not yet published formal AI safety research; the fellowship is intended as a focused transition from adjacent reliability work into rigorous, mentor-guided research.

TECHNICAL TOOLKIT

Python | C#/ .NET | SQL | PostgreSQL | AWS | Azure | Git | CI/CD | xUnit | Power BI | Excel/VBA | Time-series modelling | Regression | Backtesting | Data validation

LANGUAGES & LOCATION

English C1 | German B2 | Farsi Native | German and EU citizen | Based in Berlin | Available for full-time in-person research, subject to U.S. work-authorisation requirements